



Travel Decision Intelligence Platform

1. Project Overview

Project Name - VoyageIQ

Project Type – Travel Decision Intelligence Platform

Project Description - VoyageIQ is a web-based travel intelligence platform designed to help travelers make better travel decisions before booking their trips.

Unlike traditional travel booking websites, VoyageIQ focuses on providing travel insights such as flight information, hotel comparisons, weather forecasts, budget estimation, and travel recommendations. The platform assists users in evaluating whether a destination is suitable for travel based on multiple factors rather than directly booking flights or hotels.

The system aggregates travel-related information from external APIs and presents it through an intuitive dashboard, helping users plan their trips more effectively.

2. Project Objectives

Primary objectives

- Provide flight information from multiple destinations.
- Provide hotel information for selected destinations.
- Display destination weather forecasts.
- Estimate travel budgets.
- Generate travel recommendations.
- Store user preferences and travel history.
- Help users make informed travel decisions before booking.

Secondary objective

- Create a modern travel analytics dashboard.
- Learn and implement Flask architecture best practices.
- Integrate third-party APIs.
- Develop a scalable backend structure.
- Prepare a foundation for future AI-powered travel recommendations.

3. Technology Stack

Frontend

- HTML5
- CSS3
- Bootstrap 5
- JavaScript
- Jinja2 Templates

Backend

- Python
- Flask

Database

- PostgreSQL

APIs

- SERPAPI Google Flights API
- SERPAPI Google Hotels API
- Open-Meteo Weather API

Development Tools

- VS Code

- Git & GitHub
- pgAdmin4

4. System Architecture

VoyageIQ follows a layered architecture to maintain scalability and code maintainability.

Architecture Layers

Presentation Layer

Responsible for

- HTML Templates
- CSS styling
- JavaScript Interactions
- User Forms

Route Layer

Responsible for

- Handling Requests
- URL Routing
- Session Management
- Returning Responses

Service Layer

Responsible for

- Business Logic
- Data Processing
- API Communication
- Recommendation Calculation

Model Layer

Responsible for

- Database Operations
- SQL Queries
- Data Persistence

Utility Layer

Responsible for

- Common Function
- Validators

5.Application Modules

5.1 Application Module

Features

- User Registration
- User login
- User logout
- Session Management

Session Data

- User id
- User name
- Email

Access Control

- Dashboard
- Profile
- Save Trips
- User Preferences

5.2 Flight Search Module

Purpose: Provide flight information between selected origins and destinations

Features

- Airport Search
- Travel Class Selection
- One-way and Round-trip Support
- Passenger Selection
- Flight Comparison

API Used

SERPAPI Google Flights

- Airline
- Flight Number
- Departure Airport
- Arrival Airport
- Departure Time
- Arrival Time
- Duration
- Cabin Class
- Price

5.3 Hotel Search Module

Purpose: Provides hotel information for selected destinations.

Features

- Destination Search
- Room Selection
- Hotel Comparison
- Price Comparison

API Used

SERPAPI Google Hotels

Information Displayed

- Hotel Name
- Rating
- Review Score
- Room Type
- Amenities
- Price Per Night

5.4 Weather Module

Purpose: Displays destination weather information.

API Used

Open-meteo API

Features

- Current Temperature
- Feels Like Temperature
- Weather Condition
- Rain Probability
- Humidity
- Wind Speed
- Visibility

Addition Functionality

Weather condition are mapped to user-friendly recommendation and weather icons.

5.5 Profile Module

Purpose: Manage user profile information.

Features

- View Profile
- Update Profile
- Personal Information Storage

Stored Information

- Full Name
- Username
- Email
- Phone Number
- Country
- Profile Picture
- Account Creation Date

5.6 Dashboard Module

Purpose: Provides a centralized travel intelligence dashboard.

Features

- Quick Search
- Travel Insights
- Budget Estimation
- Travel Recommendations
- Weather Summary

Dashboard Search

The dashboard performs live API calls and does not store search results in the database.

6.Database Design

Main Tables

users

Stores user profile information.

authentication

Stores login credentials and authentication-related data.

user_preferences

Stores user travel preferences.

searches

Stores user search history.

flights

Stores retrieved flight information.

hotels

Stores retrieved hotel information.

weather_data

Stores destination weather information.

target_trips

Stores long-term travel plans.

7.Security Features

Implemented

- Session-based Authentication
- Form Validation using Flask-WTF
- Protected Routes
- Database Parameterized Queries
- Input Validation

8. User Workflow

Guest User

- Visit Homepage
- Search Flights
- Search Hotels
- Check Weather
- Register/Login

Logged-In User

- Login
- Access Dashboard
- Manage Profile
- Create Target Trips
- Search Travel Information
- View Personalized Dashboard

